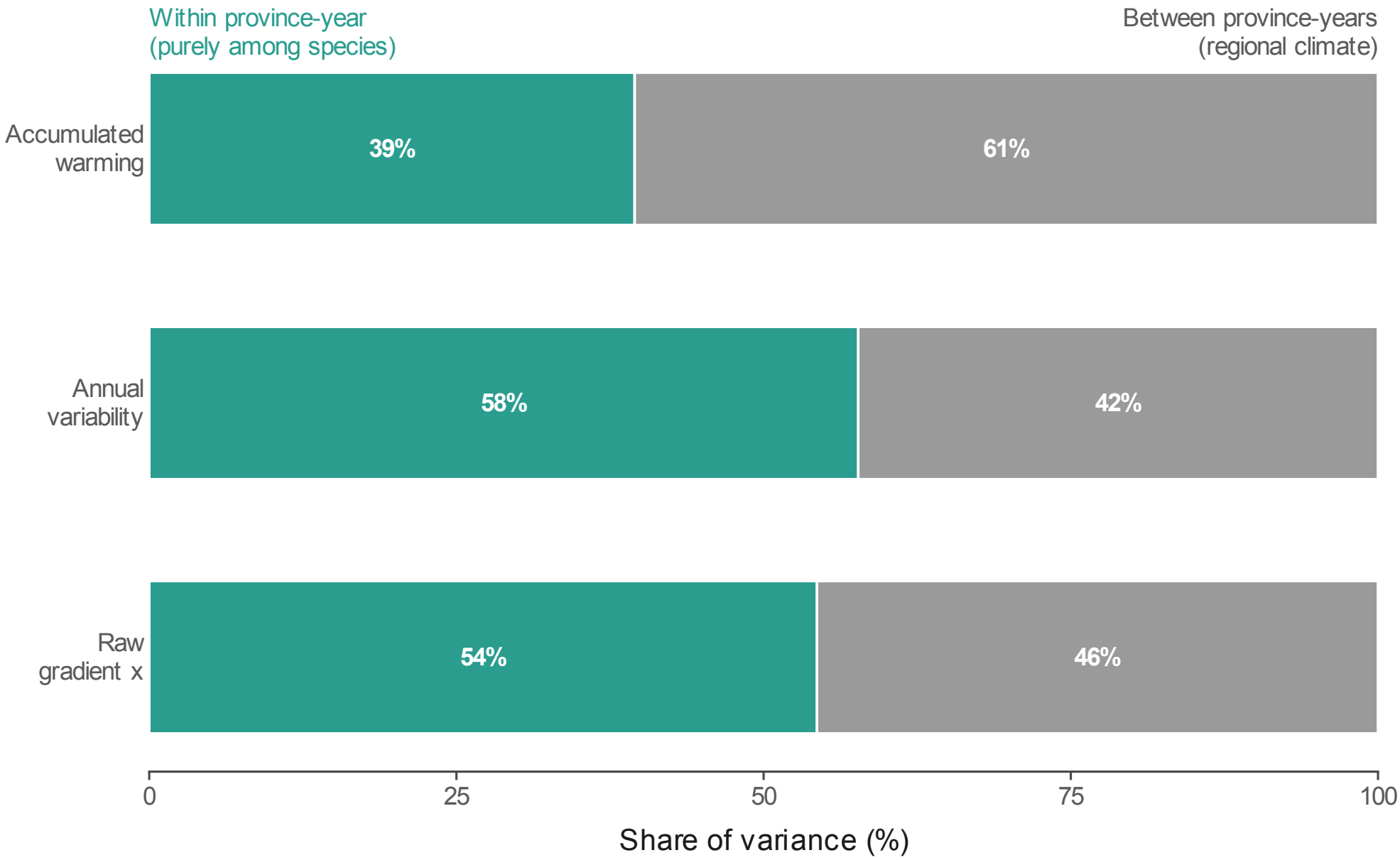Species identity carries much of the variance